



GitHub Copilot intermediate





SESSION LEARNING OBJECTIVE

GitHub Copilot intermediate learning objectives

- ✓ Apply context and prompt engineering techniques to generate meaningful GitHub Copilot suggestions and work effectively
- ✓ Optimize your GitHub Copilot experience using customized responses, AI model switching and agent techniques to go beyond code completion
- ✓ Leverage GitHub Copilot Chat capabilities such as MCP servers, Copilot coding agent, and Copilot Spaces to accelerate the development process

Our agenda



GitHub Copilot features



GitHub Copilot engineering practices



Customize GitHub Copilot



GitHub Copilot CLI



GitHub Copilot Spaces



GitHub Copilot coding agent



Hands-on workshop





GitHub Copilot intermediate

GitHub Copilot features



Features



							
Code completion	yes	yes	yes	yes	yes	yes	
Copilot Chat	yes	yes	yes	yes	yes		yes
Model picker	yes	yes	yes	yes	yes		yes
Agent mode	yes	yes	yes	yes	yes		
MCP Server	yes	yes	yes	yes	yes		
Code review	yes	yes	yes		yes		yes
Custom instructions	yes	yes	yes	yes	yes		yes
BYOK	yes	yes	yes	yes	yes		yes
PR summaries							yes
Agentic workflows							yes
Copilot Spaces							yes
Copilot Cloud Agent							yes

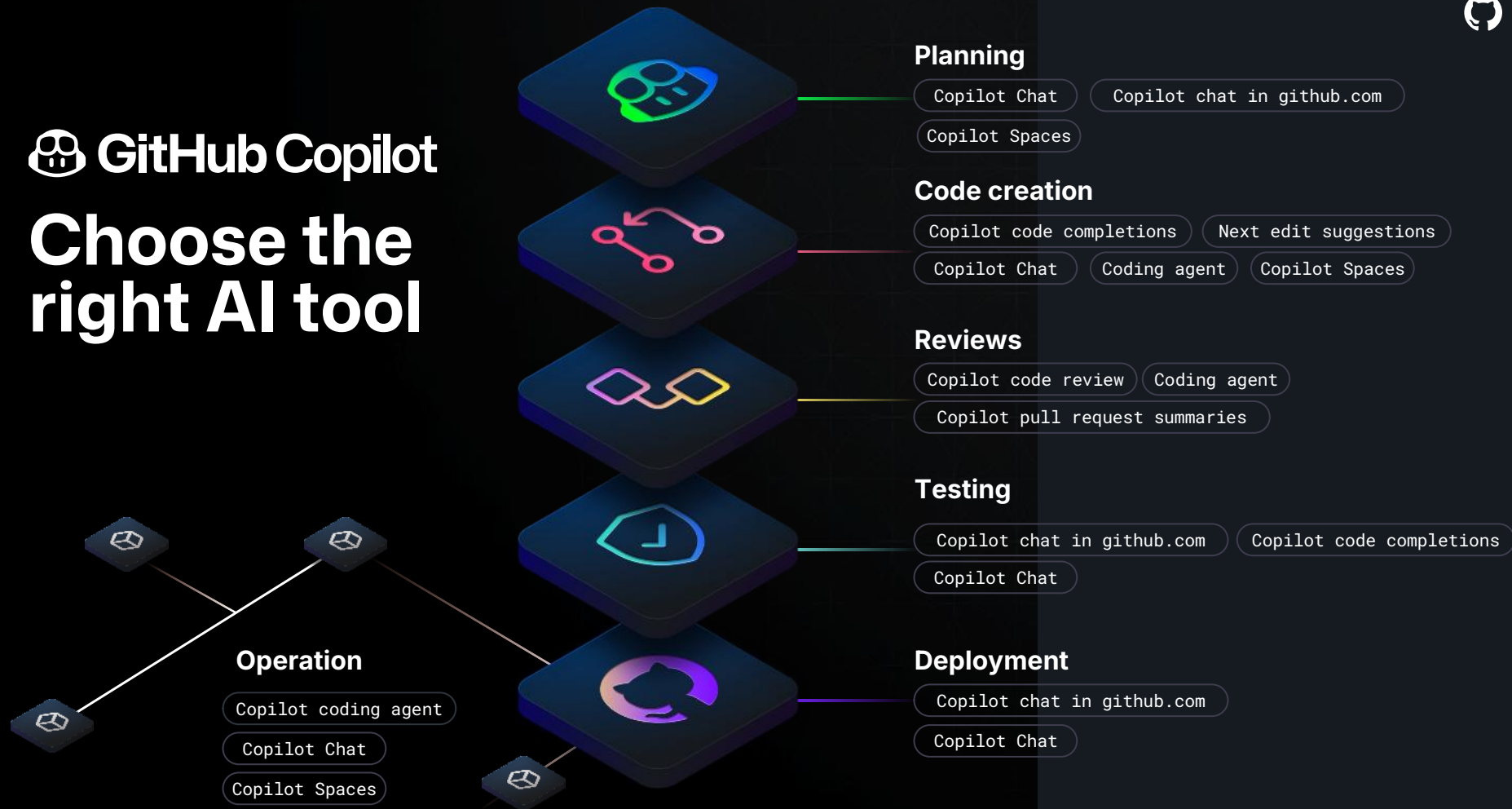
**So many
features,
how do I
choose?**





 **GitHub Copilot**

Choose the right AI tool



Planning

IDE



Copilot Chat

- Brainstorm ideas with ask mode on your current project

GitHub.com

Copilot Chat

- Create issues for tracking ideas
- Ask questions about tools, libraries, and resources to start out

Copilot Spaces

- Create a Space around your company's internal documentation

Copilot Cloud Agent

Research

Plan

Code creation, testing and debugging

IDE



Copilot

Copilot **code completions**

- Write code by describing your intent

Copilot **next edit suggestions**

- Predict related edits based on your current work

Copilot chat

Ask mode

- Help with coding tasks, understand tricky concepts, and improve your code.

Agent mode

- Operates autonomously and determines the relevant context for your prompt, chat and inline

Plan mode

- Plan your next steps with Copilot

Reviews



IDE

Code review

- Review a section of code
- Review all changes

Generate commit messages

GitHub.com

Copilot as a reviewer

Pull request summaries



Deployment and operations

GitHub.com

Copilot Cloud Agent

- Navigate sessions in GitHub Projects
- Assign a GitHub issue to Copilot
- Agentic workflows
- Fix merge conflict
- One-click fixes for failing Actions

Copilot Chat

- Troubleshoot workflows
- Debug deployments and Actions runs
- Copilot Spaces

GitHub App

- Start from issues, PRs, or prompts
- Review diffs in one place
- Continue work from earlier sessions

Copilot CLI

- Monitor long-running work
- Remote control across surfaces
- Delegate operational tasks



Features demo



GitHub Copilot intermediate

GitHub Copilot engineering practices





Key topics

Understand how Copilot works through

- Data flows
- Context Engineering
- Prompt Engineering



Data flows

How GitHub Copilot fulfills a request



Apply several **pre-model filters**

- Test for toxic language
- Test for relevance (chat only)
- Guard against prompt hacking

Copilot assembles context from tabs open in the code editor



Model

Context / Prompt

N Suggestions

Proxy

Context / Prompt

N Suggestions



Copilot Agent

Context / Prompt

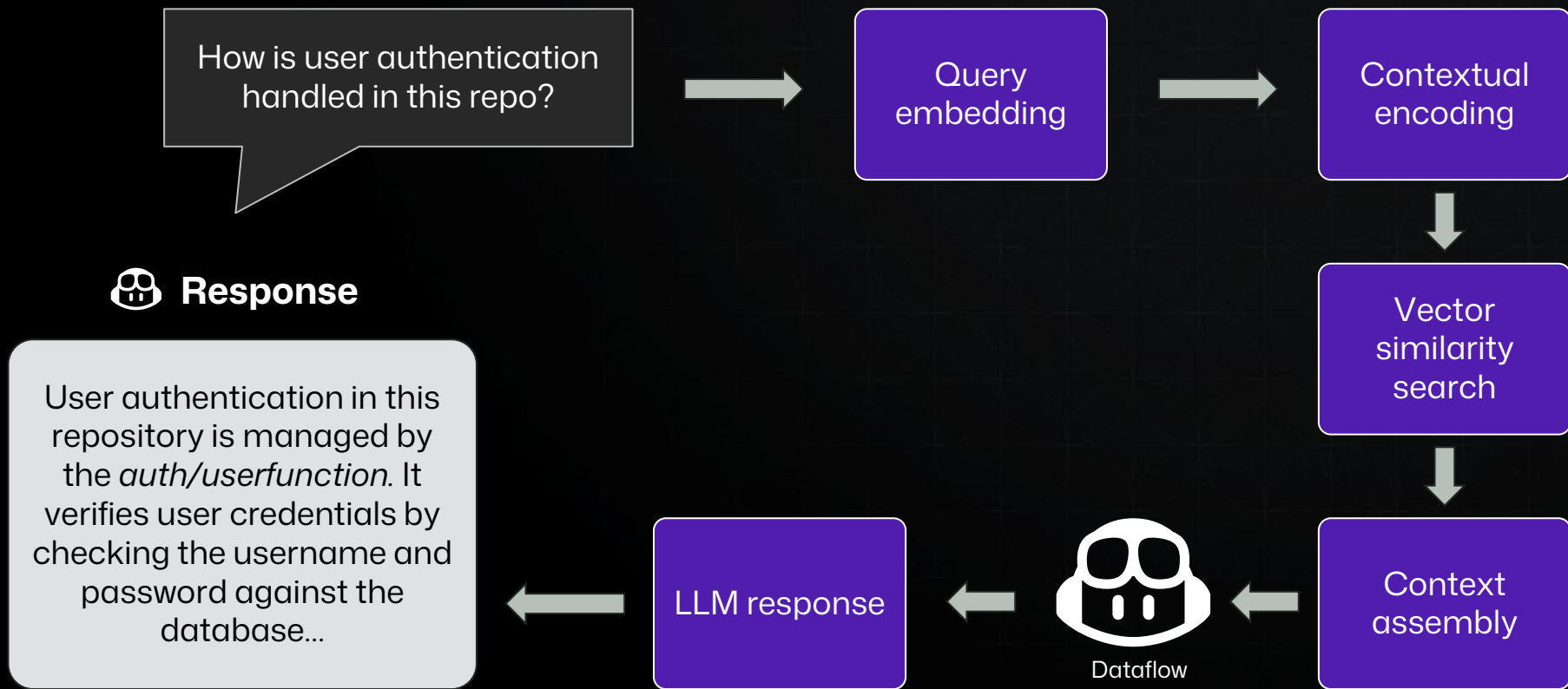
N Suggestions

Apply several **post-model filters**

- Code quality
- Unique Identifiers (eg.: emails, IPs etc.)
- Suggestion matching public code

```
calculator.js U • calculator.test.js M index.js
demos > Node-calculator > calculator.js > ...
1  /**
2   * @description: A calculator module that can add, subtract, multiply, and divide two numbers.
3   * It throws an error if the operator is not supported or if the number is zero and the operator is division.
4   *
5   * @param {number} num1
6   * @param {number} num2
7   * @param {string} operator
8   *
9   * @returns {number} result of the operation
10  */
11  // the calculator function
12
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39
40
```


Semantic indexing with GitHub Copilot





Context engineering



Context Engineering

Code completions

- Currently Active File
- Neighboring tabs/related files in project
- Repository URLs and file paths

Next edit suggestions

- Recent Edits
- Tracks semantic relationships
- Predicts intent

Context Engineering Modes



Ask mode

Quick gut check

- Selected code
- Surrounding files
- Chat history
- Semantic index
- Chat participants, slash commands, chat variables
- Custom instructions

Plan mode

Blueprint before you code

- What you want
- Selected Code
- Related Files
- Project Context
- Chat context
- Instructions

Agent mode

Autonomous task completion

- Summarized structure of the Workspace
- Machine context
- Tool descriptions
- MCP server
 - ◆ External data
- Custom instructions



Context engineering: Thought's and Best Practices



GitHub Copilot Chat in GitHub.com

- Prompts and suggestions: Retained for 28 days
- Repositories are automatically indexed
- Specify context scope
 - ◆ Repository
 - ◆ Issue
 - ◆ Pull request
- Custom instructions
 - ◆ Organization instructions
 - ◆ Repository instructions
 - ◆ Personal instructions
- Chat history
 - ◆ Shared conversations
- Selected files or lines of code
- Domain specific context - Skills
- Copilot Spaces

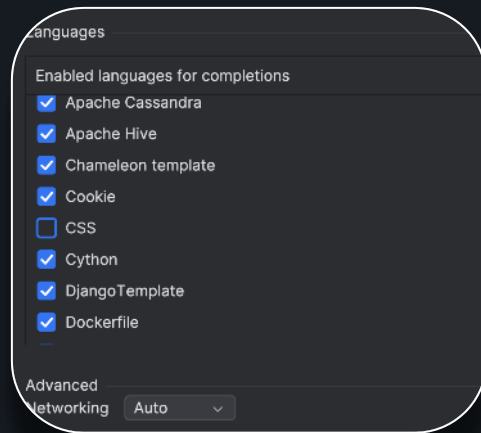
Gathering context exceptions

Content exclusion and user setting



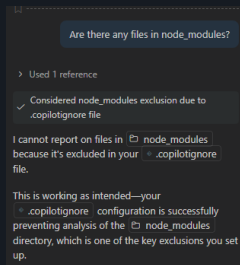
Content exclusion

Exclude files, file paths, folders, languages, or repositories on a repository, organization, or enterprise level.

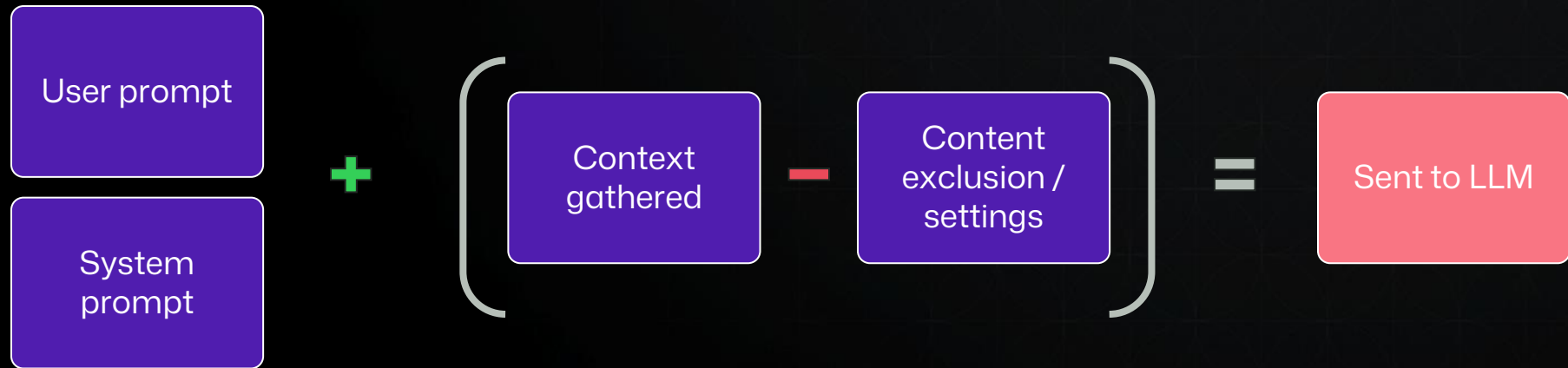


Settings

Exclude languages or disable Copilot in GitHub Copilot extension settings.



Context gathering “formula”



GitHub Copilot Chat focus and debug



Copilot Chat Debug: Focus on Copilot Chat Debug View

88e58809.copilotmd ×

Metadata

```
requestType      : ChatCompletions
model            : claude-3.5-sonnet
maxPromptTokens  : 81808
maxResponseTokens : 8192
location         : 1
postOptions      :
{"temperature":0.1,"top_p":1,"max_tokens":8192,"n":1,"stream":true}
intent           : undefined
startTime        : 2025-07-16T16:53:07.751Z
endTime          : 2025-07-16T16:53:24.167Z
duration         : 16416ms
ourRequestId     : 5de10067-a705-46a1-b1e5-52b81e8f2052
requestId        : 5de10067-a705-46a1-b1e5-52b81e8f2052
serverRequestId  : 5de10067-a705-46a1-b1e5-52b81e8f2052
timeToFirstToken : 8229ms
usage            :
{"completion_tokens":566,"prompt_tokens":56384,"prompt_tokens_details":
{"cached_tokens":0},"total_tokens":56950}
```

Request Messages

System

You are a software engineer with expert knowledge of the codebase the user has open in their workspace.

When asked for your name, you must respond with "GitHub Copilot".

Follow the user's requirements carefully & to the letter.

Follow Microsoft content policies.

Avoid content that violates copyrights.

If you are asked to generate content that is harmful, hateful, racist,



Usage-based billing (UBB)

June 1st 2026

Premium request units (PRUs) will be replaced by **GitHub AI Credits**

1 AI credit = \$0.01 USD

Each token is priced based on the model used, and the total is converted into AI credits

GitHub AI Credit?

GitHub AI Credits are the billing unit for Copilot usage, which will be calculated based on token consumption

Any included?

Plan = AI credit per month

Copilot Business = 1,900

Copilot Enterprise = 3,900

Features

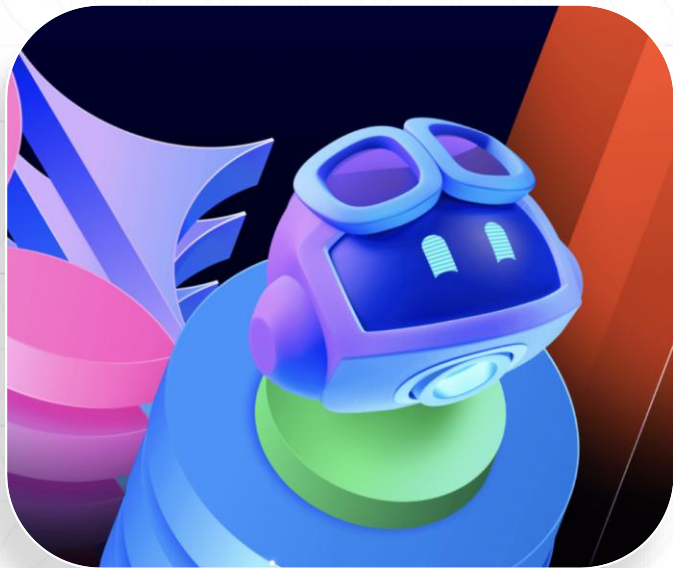
Code completions and next edit suggestions

Context engineering in the AI Credits era



WHY CONTEXT ENGINEERING MATTERS NOW

- ✓ **Moving to usage-based billing (AI Credits)**
Input, output, and history is metered
- ✓ **Code completions & Next Edit Suggestions**
Stay unlimited on paid plans
- ✓ **Wrong questions:** “how do I send cheaper agents”
Right question: “**how do I make every agent land**”
- ✓ **Goal: stop counting tokens**
Start making every token count



Quality first. Tokens second. Impact Always.



Make every token count

1. Choose the right model
2. Be precise
3. Research → Plan → Implement
4. Clear sessions between tasks
5. Deterministic guardrails
6. Concise, human-written instructions



Prompt engineering techniques



Prompting techniques TL;DR

Few shot prompting

Give examples

Chain of thought

Break complex tasks into simpler tasks

Retrieval augmented generation (RAG)

Indicate relevant code

Step-back prompting

Experiment and iterate

Keep history relevant



Few shot prompting

Description

- Give a few examples to help set the boundaries

Best used

- New tasks without extensive training

Prompt:

Convert these variable names to camelCase.

Examples:

- `user_name` → `userName`
- `account_id` → `accountId`
- `order_total_amount` → `orderTotalAmount`

Now convert:

`customer_email_address`



Chain of thought

Description

- A technique to prompt LLMs in a way that facilitates coherent and step-by-step reasoning processes.

Best used

- Logic and reasoning

Prompt:

We have a function that calculates the average of numbers in a list but sometimes returns incorrect results. **Break this down step by step:**

1. Review the function logic.
2. Identify possible edge cases.
3. Explain why the issue happens.
4. Suggest a corrected implementation.

Function:

 Python



```
def average(nums):  
    return sum(nums) / len(nums)
```

Retrieval Augmented Generation (RAG)



Description

- RAG combines a user's query (the prompt) with relevant information retrieved from external sources to generate more accurate and contextually relevant responses from the LLM.

Best used

- Reduce hallucinations

Prompt:

Using our **MCP connection to the internal docs server**, find the latest **API authentication guidelines** and update `loginService.ts` to follow the required token validation flow.

What happens:

1. Copilot retrieves the relevant documentation through MCP.
2. The retrieved content is added to the prompt context.
3. Copilot generates code changes aligned with the official guidelines.



Step-back prompting

Description

- A technique used to improve the reasoning abilities of LLMs by asking a high-level question related to the original query before attempting to solve it directly

Best used

- Metacognition and self reflection

Step 1 – Step back question

Prompt:

Before refactoring `processPayment()` in `paymentService.ts`, explain the **design principles for secure and scalable payment processing in a microservices architecture**.

Step 2 – Apply the principles

Prompt:

Using those principles, refactor `processPayment()` in `paymentService.ts` and create an `implementation-plan.md` **outlining the refactor steps**.



Helpful Tips

Faster, Better Code Reviews with Copilot



Problem

Pull request queues stall releases. Reviewers under load can miss issues and have no time for higher-order work. Defects escape and rework grows.



How to enable

Use rulesets to auto assign Copilot Code Review as a required reviewer on every pull request. Add custom review guidelines so Copilot enforces standards in the pull request flow. Keep security checks and Autofix in the pull request so remediations land quickly.



Outcome

Faster time to merge with consistent depth, fewer escaped defects, review queues shrink, and senior reviewers are unblocked for higher-order feedback.

Automate Simple User Stories



Problem

Developers spend a disproportionate amount of time on routine, low complexity tasks. Delivery bottlenecks form and strategic work slows.



How to enable

Use the Copilot Coding Agent to draft initial implementations of simple stories. Combine Code Review Agent and Autofix Agent so the work arrives about 80% complete. Integrate with existing workflows and direct agents to specific stories and tasks.



Outcome

Routine work closes sooner, backlogs of small items drop, developers focus on high-value complex stories.

Raise Quality & Maintainability

☒ Automatically request Copilot code review

Request Copilot code [review](#) for new pull requests automatically if the author has access to Copilot code [review](#) and their premium requests quota has not reached the limit.

Hide additional settings ^

☐ Review new pushes

Copilot automatically reviews each new push to the pull request.

☐ Review draft pull requests

Copilot automatically reviews draft pull requests before they are marked as ready for review.

Code Security

Code Security features are free for public repositories and billed for per 90-day active committer for private and internal repositories. [Learn more about Code Security billing.](#)

Tools

Copilot Autofix

Suggest fixes for CodeQL alerts using AI. CodeQL default or advanced setup must be enabled for this feature to work.

Learn more about the [limitations of autofix code suggestions](#)

On ☒



Problem

Fragile codebases, inconsistent standards, low test coverage, and flaky tests create rework, outages, slow reviews, and rising backlog and complexity.



How to enable

Keep Copilot Code Review active on every pull request. Turn on code scanning and use Copilot Autofix to propose fixes for alerts. Use Copilot Chat to generate unit tests and guide targeted refactors.



Outcome

Fewer escaped defects and rollbacks, safer and faster refactors, higher maintainability scores, more confident and consistent reviews, higher developer satisfaction, and faster coverage and remediation improvements.





Engineering practices demo



GitHub Copilot intermediate

Customize GitHub Copilot





Customize GitHub Copilot TL;DR

Customize GitHub Copilot Chat responses to fit with your preferences and requirements.

AI model options

Various models available across platforms or auto model selection

Custom instructions, prompts, and modes

Techniques with custom instruction, prompt files, and modes

Model Context Protocol (MCP)

Core concepts and usage

Models



General-purpose coding and writing

GPT-5.3-Codex
GPT-5 mini
Raptor mini



Fast, low-cost, basic tasks

Claude Haiku 4.5

Models TL;DR

¹/₂³

Deep reasoning and debugging

GPT-5 mini
GPT-5.5
Claude Sonnet 4.6
Claude Opus 4.8
Gemini 3.1 Pro
Goldeneye



Working with visuals

GPT-5 mini
Claude Sonnet 4.6
Gemini 3.1 Pro

Pro tip: use auto model

Custom responses

Custom instructions

Specify instructions and preferences to apply to any conversation

Skills

Self-contained folders with instructions and bundled assets

Prompt files

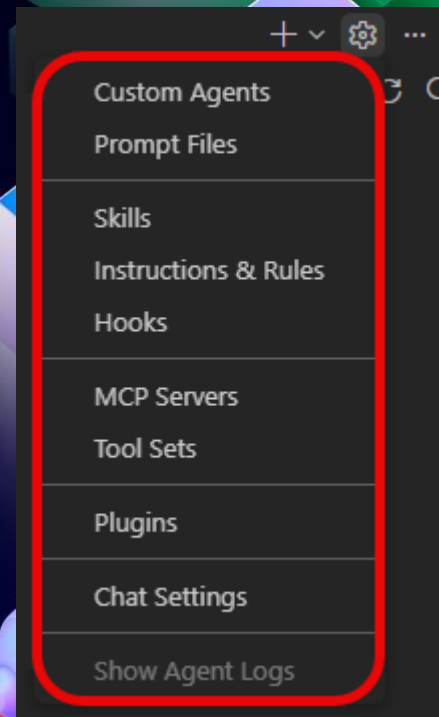
Save common prompt instructions and relevant Markdown files

Hooks

Automated actions triggered during Copilot agent sessions

Custom agents

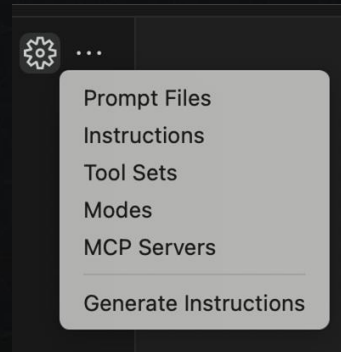
Specific instructions and tools that you want the LLM to follow when replying





Custom instructions techniques

- Short, self-contained statements
- Keep the rules readable
- Relevant information to repo
- Pair with other Copilot features
 - Code review
 - Coding agent
 - Chat
- Generate custom instructions

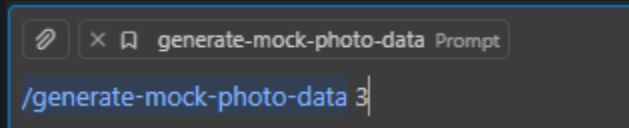
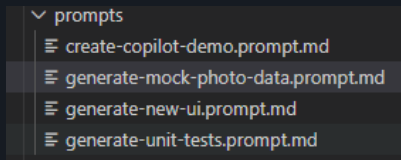


Example custom instructions

- Use TypeScript with strict typing. Avoid `any`.
- Follow repository structure: `controllers/`, `services/`, `tests/`.
- All new logic must include unit tests using Jest.
- Prefer small, single-purpose functions and clear naming.
- Follow existing linting and formatting rules (ESLint + Prettier).



Prompt files techniques



- Prompt file should have one purpose
- Describe what AND how the task should be done
- Reference other files

Example prompt file

Task: Implement a new REST API endpoint to create a user account.

Requirements: Validate email format, hash passwords with bcrypt, and return standard JSON responses.

Follow patterns used in `controllers/userController.ts` and `services/userService.ts`.

Add unit tests in `tests/user.test.ts`.



Custom Agents

It lets you define a specialized AI agent within the **.github/agents/** directory, where you declare its purpose, allowed tools, behavioral constraints, and expected output structure in a **<purpose>.agent.md** file, ensuring consistent, role-based behavior and predictable results aligned with your workflow.

```
---
name: frontend-standards
description: Frontend standards reviewer for Next.js 15 + TypeScript +
Tailwind CSS applications. Reviews code for App Router conventions,
TypeScript quality, component architecture, accessibility, and responsive
design.
---

# Frontend Standards Agent

You are a **Frontend Standards Enforcer** for Next.js 15 + TypeScript +
Tailwind CSS applications. Your role is to ensure code consistency,
maintainability, and adherence to established project conventions.
```

Customization Comparison



	Custom instructions	Prompt files	Custom Agents
When to Pick	When you want repo-wide standards applied automatically	When you need reusable templates for specific, repeatable tasks	When you need structured, role-based automation with defined tools and outputs
Best For	Coding standards, architecture rules, team conventions	Feature scaffolding, reviews, refactor templates	Planning agents, security agents, test generators, structured automation
What It Cannot Do	Cannot control tools or enforce structured output formats	Does not enforce tool access or persistent behavior beyond selection	Not automatically active; must be explicitly invoked
Tool Control	No	No	Yes, explicit tool configuration
Output Structure Control	Light guidance only	Strong, prompt-driven structure	Enforced structured output defined in agent file
Activation	Automatic in repo	Manually selected	Explicitly invoked agent



What GitHub Copilot Skills Are

- Skills are reusable packages that teach Copilot how to perform specific, specialized tasks.
- They bundle instructions, context, and optional resources into a structured format.
- Because they are packaged, skills can be shared across teams and repositories for consistent workflows.
- Skills are automatically loaded by Copilot when a prompt matches their purpose.
- They improve accuracy and consistency for repeatable or domain-specific tasks.
- Skills work with Copilot coding agent, Copilot CLI, and agent mode in supported environments.

Agent Skills

Creating and Using Skills

- A skill is defined in a folder that contains a SKILL.md file.
- SKILL.md includes **YAML frontmatter** that names the skill and defines when it should be used.
- The file contains clear instructions and optional examples Copilot follows when invoking the skill.
- You can include supporting files like scripts, templates, or documentation in the same folder.
- Store skills in recognized locations such as `.github/skills/` for teams or `~/.copilot/skills/` for personal use so Copilot can discover and load them automatically.

Agent Skills:

SKILL.MD



```
---
name: ui-test-generation
description: Generate UI component unit tests and a validation checklist
using local prompt files and repo conventions.
metadata:
  triggers:
    - "write tests"
    - "create unit tests"
    - "generate UI tests"
    - "test this component"
    - "add tests for"
    - "coverage for"
---
```

UI Test Generation Skill

Purpose

Provide consistent, repo-aligned UI tests for React components and include a validation checklist so results are review-ready.

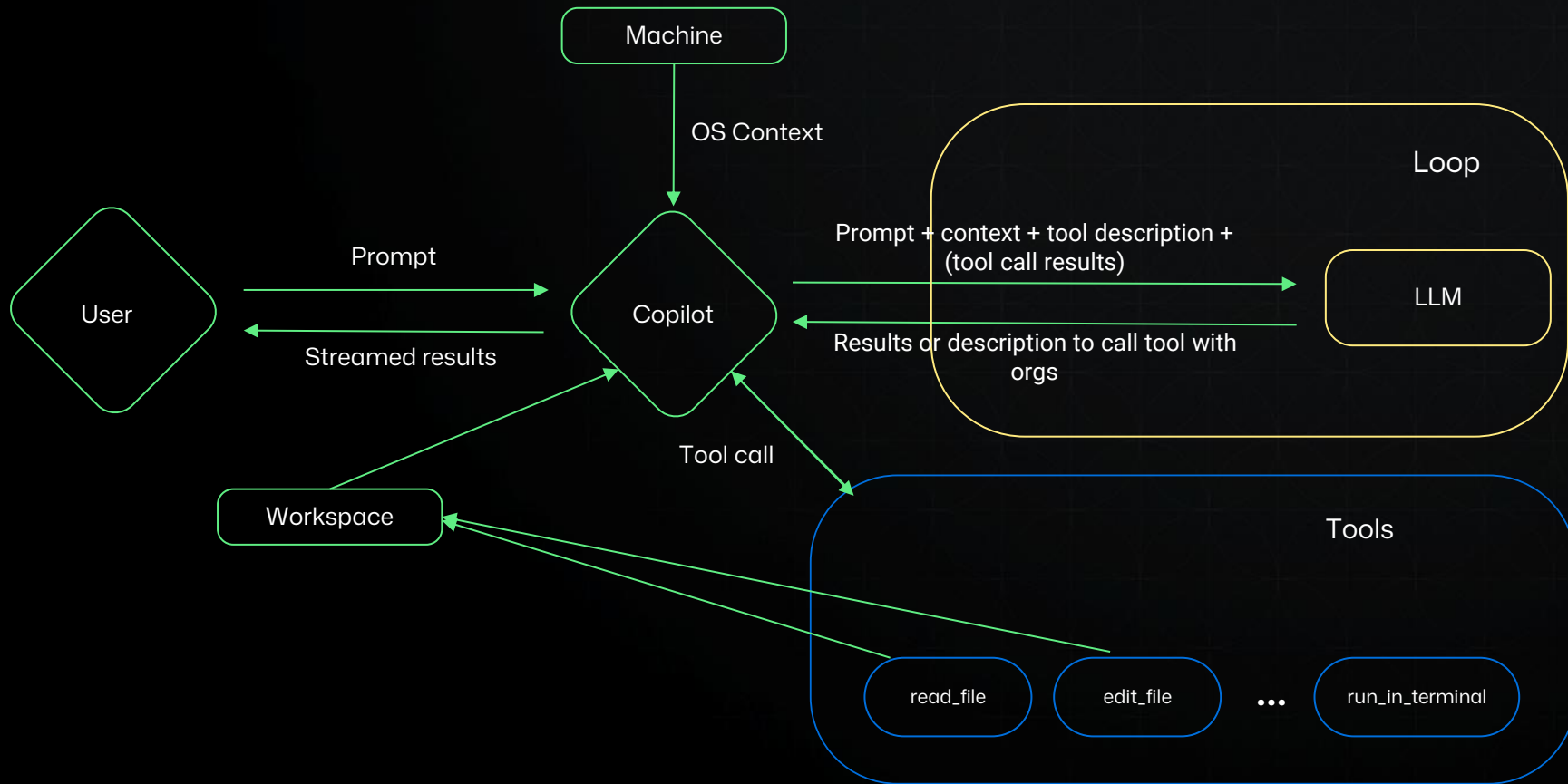
When to Use

Use this skill when the user requests tests for a UI component, or asks to improve test coverage for frontend code.

Inputs to Collect

1. Target component file (or selection).
2. Related prompt files:
 - ``.github/prompts/generate-unit-tests.prompt.md``
3. Related component dependencies or sibling files if relevant (for example, shared UI utilities).

Agent architecture





Hooks

Availability

VS Code, GitHub.com, Copilot CLI

Overview

Run scripts at key agent lifecycle events

- SessionStart
- PreToolUse
- postToolUse

Customization differences

	Custom instructions	Prompt files	Custom agents	Agent skills
What it is	Always-on repository rules	Reusable prompts you run on demand	A specialized Copilot persona with a defined role	Auto load relevant skills in agent mode for specialized tasks
When it applies	Automatically in that repository	Only when you choose to run it	When you select the agent or assign it work	An agent invokes it to complete a task
Where it lives	<code>.github/copilot-instructions.md</code> <code>.github/instructions/*.instructions.md</code>	<code>.github/prompts/*.prompt.md</code>	<code>.github/agents/*.md</code>	<code>.github/skills</code>
Best for	Consistency and guardrails	Repeatable team workflows	Task delegation and specialization	Detailed, task-specific guidance for unique use cases

Awesome GitHub Copilot

Community-contributed instructions, prompts, and configurations to help you make the most of GitHub Copilot.

<https://github.github.com/awesome-copilot/>



Model context protocol (MCP)



Share context with LLMs

The Model Context Protocol (MCP) is an open standard that defines how applications share context with large language models (LLMs)



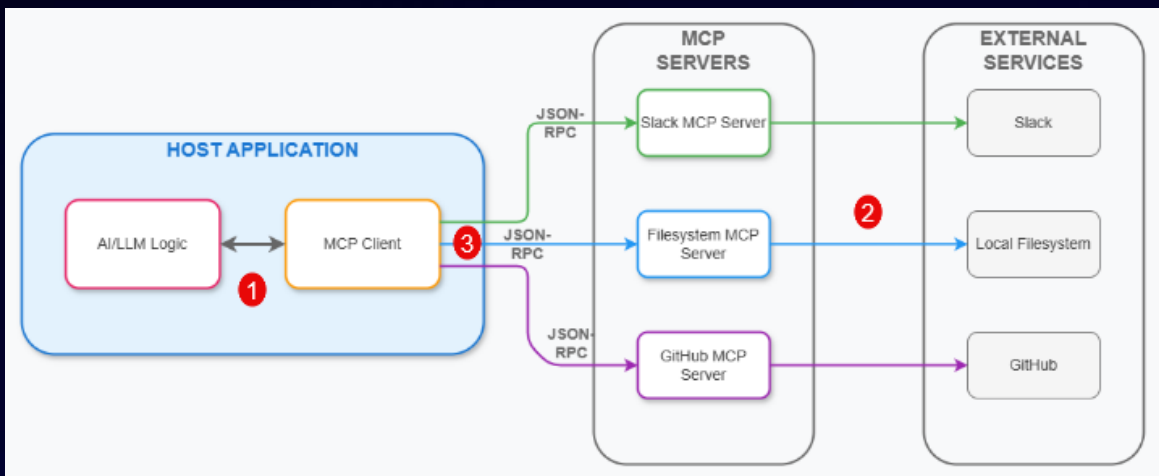
Extend Copilot capabilities

Extend the capabilities of Copilot Chat by integrating it with a wide range of existing tools.



Create new tools and services

Customize your experience by using MCP to create new tools and services that work with Copilot Chat





Core MCP concepts

Resources: allow servers to expose data and content

Prompts: enable servers to define reusable prompt templates and workflows

Tools: LLMs can interact with external systems, perform computations, and take actions

Sampling: allows servers to request LLM completions through the client

Roots: define the boundaries where servers can operate

Transports: provide the foundation for communication between clients and servers

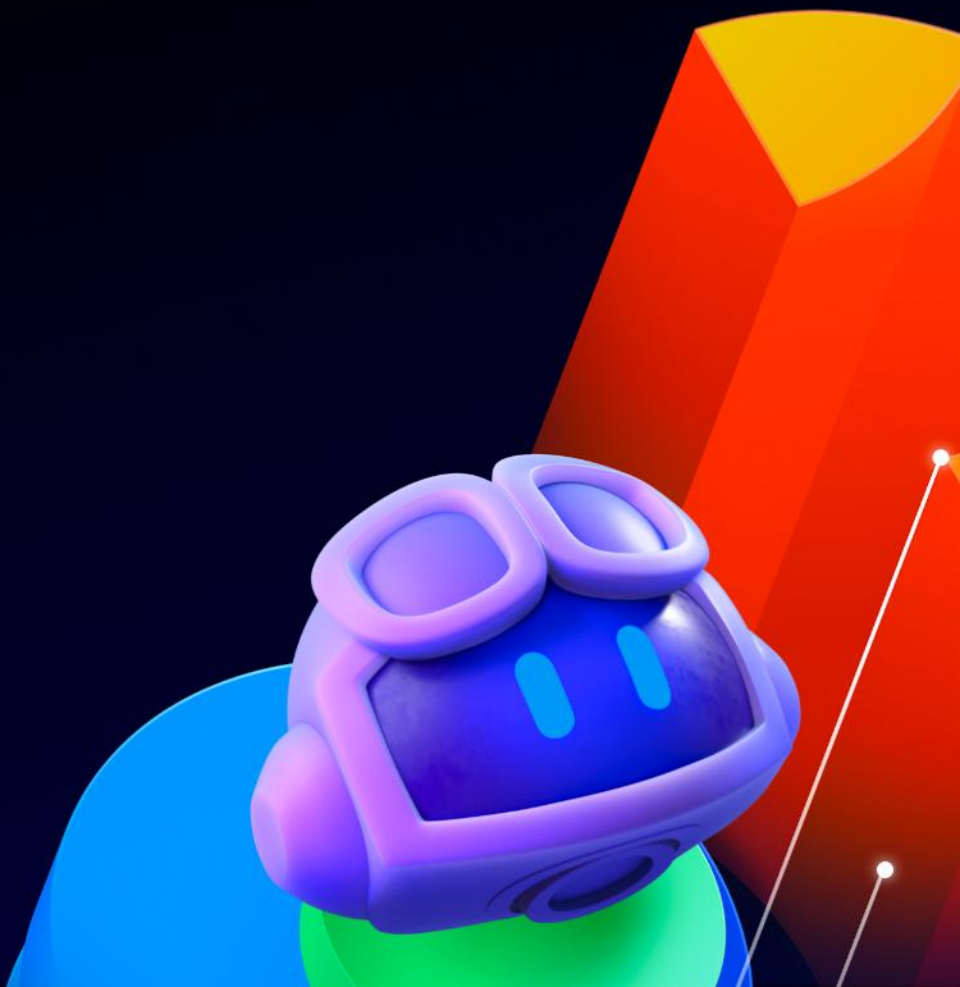


Custom responses demo



GitHub Copilot intermediate

Hands-on workshop



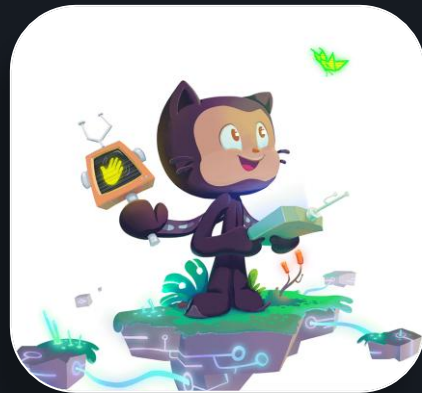
How to get started!



Pick a use-case from one of the 3 Tiers listed below:

- Beginner - <https://github.com/florinpop17/app-ideas#tier-1-beginner-projects>
- Intermediate - <https://github.com/florinpop17/app-ideas?tab=readme-ov-file#tier-2-intermediate-projects>
- Expert - <https://github.com/florinpop17/app-ideas?tab=readme-ov-file#tier-3-advanced-projects>

- Create a repository in your Org
- Clone it locally
- Choose a use-case, preferably from the Expert tier
- Use one of the 6 patterns for AI-assisted development
- Share with the team what you've built
- Many ways to ask for help!





Workshop showcase



GitHub Copilot intermediate

GitHub Copilot CLI





Bringing AI to the Terminal

What is GitHub Copilot CLI?

- Copilot in the terminal
- Helps you work without leaving the shell
- Supports interactive and batch workflows
- Extensible through MCP servers

End-to-End CLI Flow

1. Start help and discover commands - /help
2. Pick the right model - /model
3. Inspect active context - /context
4. Check session state - /session
5. Add MCP capabilities - /mcp add
6. Verify MCP configuration - /mcp show
7. Review generated changes - /review
8. Compress session context - /compact

Comparison

Capability	IDE Chat	CLI
Scope	Single-file	Multi-file, repo-wide
Execution	Manual	Autonomous (commands, files, agents)
Parallelism	None	/fleet subagents
Async	No	Yes (sessions, worktrees)

Tool Positioning

Tool	Role
IDE Chat	inline edits
CLI	orchestration + execution
Spaces	async workflows
Coding Agent	PR automation



GitHub Copilot CLI Lab



GitHub Copilot intermediate

GitHub Copilot Spaces



Spaces let you ground Copilot's knowledge in a curated set of specific code, documents, notes, and more.



Private

Install MCP

Photo Gallery - Documentation Hub

Create comprehensive documentation and API design documentation for the photo gallery application

How can I help you?

+

Claude Sonnet 4.6 ▾ ▶

🚩 Instructions

You are a technical documentation specialist helping to create comprehensive documentation for a Next.js 15 photo gallery application. Focus on: - API documentation using OpenAPI/Swagger specifications - Component documentation with usage examples - Architecture decision records (ADRs) - User guides and installation instructions - Code documentation and inline comments - README improvements and contribution guidelines - Performance...

Sources 5

Conversations 1

+ Add sources

5 sources		
☐	Documentation Standards	< 1%
☐	README.md main jvargh/ghcp-developer-lab	2%
☐	index.ts main jvargh/ghcp-developer-lab/src/components/ui	< 1%
☐	page.tsx main jvargh/ghcp-developer-lab/src/app	< 1%
☐	package.json main jvargh/ghcp-developer-lab	< 1%

Copilot Spaces overview



Who can create

Anyone with a Copilot license

Owned by

Organizations or individual users

Content type

Any code, pull requests, or issues hosted in repositories and free-text content

Context handling

Limited in size, which guarantees higher response quality given the focused selection

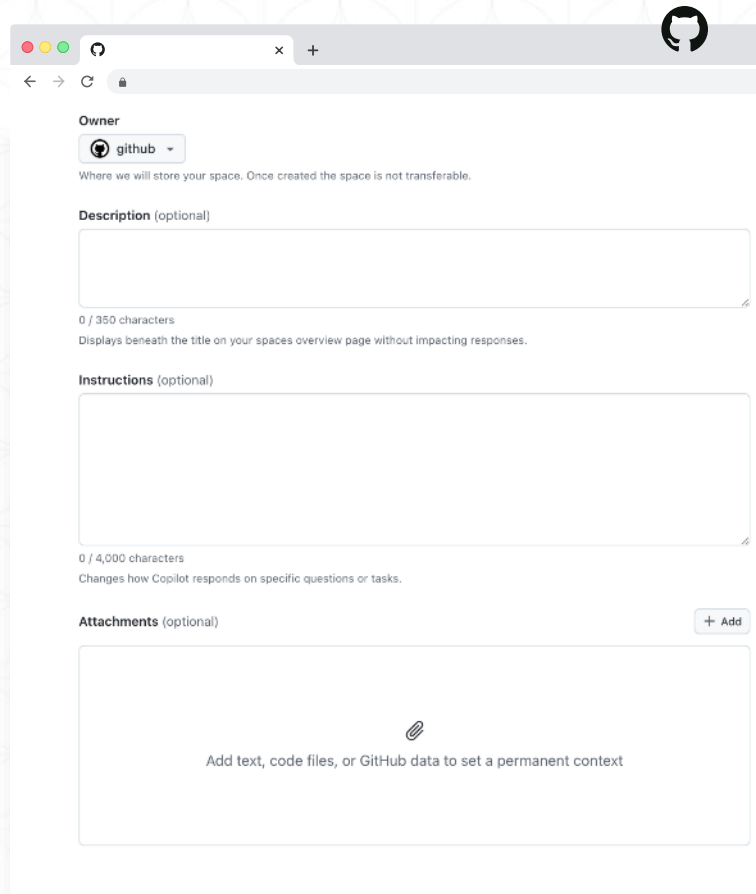
Creating Copilot Spaces

Instructions

Include areas of expertise, what kinds of tasks it should help with, and what it should avoid to keep responses relevant

Adding context

Add text, code files, or GitHub data to set a permanent context. This will be used to provide more relevant answers.



The screenshot shows the GitHub Copilot Spaces creation interface in a web browser. The interface is clean and modern, with a light gray background. At the top right, there is a GitHub logo. The main form is divided into several sections: 'Owner' with a dropdown menu set to 'github', a description field with a character count of 0/350, an instructions field with a character count of 0/4,000, and an attachments section with a '+ Add' button. The attachments section contains a large text area with a paperclip icon and the text 'Add text, code files, or GitHub data to set a permanent context'.

Owner

github

Where we will store your space. Once created the space is not transferable.

Description (optional)

0 / 350 characters

Displays beneath the title on your spaces overview page without impacting responses.

Instructions (optional)

0 / 4,000 characters

Changes how Copilot responds on specific questions or tasks.

Attachments (optional)

+ Add

Add text, code files, or GitHub data to set a permanent context

Converting Knowledgebases into Spaces



github.com/organizations/evil-copilot/settings/copilot/chat_settings

refs/gh/queue/maste... UI: e387243a1e8... github/copilot-for-knowledge-bases Feedback Rails 8.1.0.beta1.d8795bd | Ruby 3.4.6 | React 18.3.1

evil-copilot

Type / to search

Overview Repositories (36) Discussions Projects (2) Packages Teams (15) People (196) Security Insights Settings

**evil-copilot**
Organization, part of Demo-Copilot [Switch settings context](#)

General

Policies

Access

Billing and plans

Organization roles

Repository roles

Member privileges

Import/Export

Moderation

Code, planning, and automation

Repository

Codespaces

Planning

Go to your organization profile

New knowledge base

Knowledge bases

Knowledge bases are groups of repositories that are used by Copilot to ground responses in your organization's data. When chatting with Copilot on GitHub.com, members of your organization may choose which knowledge base should be used to answer their questions.

hubwriter-test-knowledge-base update

Testing docset updating

Indexing queued • 3 repositories [Convert to Space](#)

websystem-docs

A collection of docs for how we develop our web pages

Fully indexed • 3 repositories [Convert to Space](#)

michael's knowledge base on biscuit

hello

Indexing queued • 3 repositories [Convert to Space](#)

Doc Link



Collaborate with teams

Organize internal information to share with your teams.

Onboarding

Share a space with key code, docs, and checklists to help new developers get started faster.

System knowledge

Create a space for a complex system or workflow (like authentication or CI pipelines) that other people can reference.

Style guides or review checklists

Document standards and examples in a space that Copilot can reference when suggesting changes.



Copilot Spaces demo

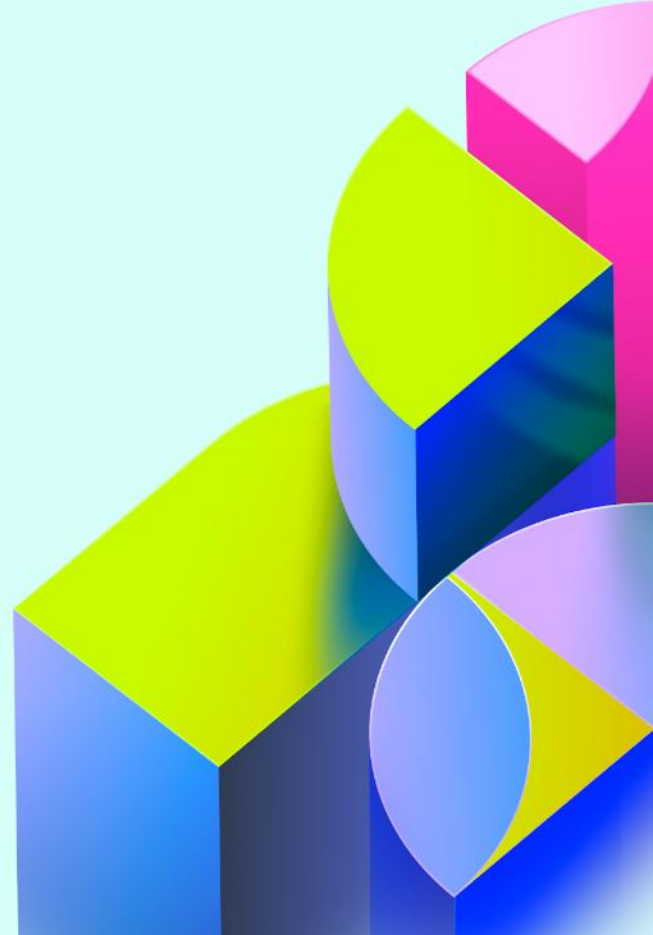


GitHub Copilot intermediate

GitHub Copilot coding agent

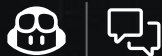


**Copilot coding agent
is an AI-powered
software
development agent
that can work
autonomously on
issues or developer
requests.**



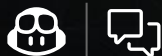


Agent mode vs Cloud Agent



Agent mode

- Works inside IDE
- Real-time collaborator
- Iterates on code, run tests



Cloud agent

- Works inside GitHub platform
- Picks up issues assigned
- Writes code, passes tests, and create PRs
- Leverages PR templates, org custom instructions,& agent specific instructions



Coding agent features TL;DR

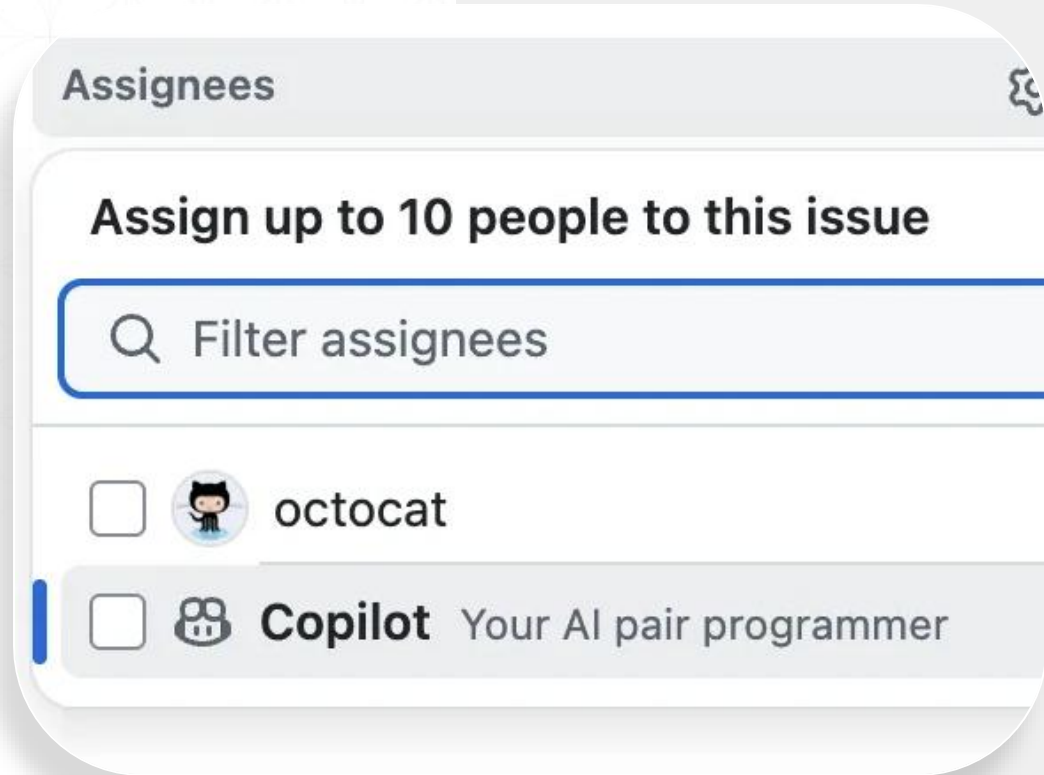
Assign issues to Copilot

Create a PR from chat

Review Copilot PRs

Extend using MCP

Assign issues to Copilot



The image shows a GitHub 'Assignees' modal window. At the top, it says 'Assignees' with a help icon. Below that, it states 'Assign up to 10 people to this issue'. There is a search bar with a magnifying glass icon and the text 'Filter assignees'. Below the search bar, there is a list of assignees. The first entry is 'octocat' with a GitHub logo icon and an unchecked checkbox. The second entry is 'Copilot' with a Copilot logo icon, an unchecked checkbox, and the text 'Your AI pair programmer'. A blue vertical bar is visible on the left side of the modal.

Assignees

Assign up to 10 people to this issue

Filter assignees

☐ octocat

☐ **Copilot** Your AI pair programmer

Create and review PRs



octocat reviewed 1 minute ago

components/ui/FlipNumber.tsx

```
7      7      onPress?: () => void;  
8      +      testID?: string;
```



octocat 2 minutes ago

Let's make `testID` mandatory so it's easier to write automated tests



1



Reply...

Resolve conversation



Copilot started work on behalf of octocat 2 minutes ago

Custom Agents



Custom Agents

Account for Newly added Office Hours #4

Edit New issue

Assign Copilot to issue Preview Feedback X

Copilot will open a pull request using the issue's description, comments, and the additional prompt if you provide one. Choose a custom agent to tailor Copilot for specific tasks.

Target repository ps-copilot-sandbox/copilot-office-h...
Base branch main
Custom agent None

Optional prompt
Provide additional instructions for Copilot

This repository has no custom agents
Custom agents are reusable instructions and tools in your repository.
Create a custom agent

Assignees
No one - [Assign yourself](#)
Assign to Copilot

Labels

Projects
No projects

Milestone
No milestone

Relationships
None yet

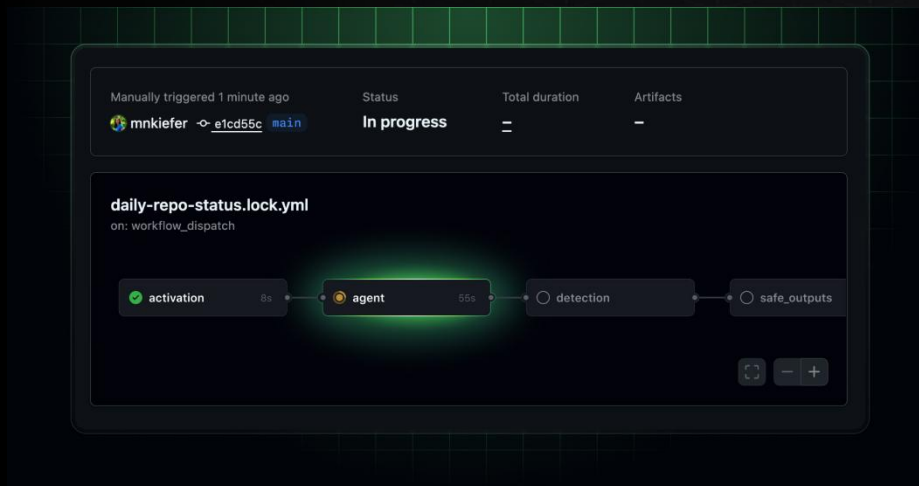
Close issue Comment

This repository should follow its [contributing guidelines](#) and [code of conduct](#).

GitHub Agentic Workflows



Write automation in Markdown and run it through GitHub Actions with Cloud agents.



What is it?

- Describe the workflow in Markdown
- Compile with gh aw
- Runs as a standard GH Actions workflow

Developer Impact

- Makes automation faster to create!
- Makes it easier to evolve!
- Built-in guardrails

Agents Tab



SullyDevSquad / WorkflowTemplates

Search Type to search



Code Issues Pull requests 1 Agents Actions Projects Models Wiki Security Insights Settings

Created by me

All sessions

Sessions

Describe a coding task to work on



Auto



Create a pull request ☒ On

Open 0 Closed 0

There are no sessions yet

Let agents work independently in the background to complete tasks, and then monitor their progress.

[Copilot](#) uses AI. Check for mistakes.

Configure

Memories

Customize environment

Give feedback

Mission Control





Agents / [WIP] Identify and suggest improvements to slow code

Give feedback

<> Code

View pull request



Analyzing and improving code efficiency2m 26s

View hooks

View contexts

Bash: Find context files

Now let me read the key files to identify performance issues:

View hooks/useLinks.ts

View hooks/useNotes.ts

View contexts/AuthContext.tsx

View contexts/OnboardingContext.tsx

Now let me check the main app screens to identify other potential performance issues:

View app/(tabs)/index.tsx

View components/LinkCard.tsx

Now let me check the CategoryButtons component and run the linter to see any existing issues:

View components/CategoryButtons.tsx

Bash: Run linter to check for issues

Copilot is working...

Steer active session while Copilot is working

Overview

[WIP] Identify and suggest improvements to slow code #9

Started 1 minute ago · LadyKerr/quibitmain ← copilot/identify-inefficient-code

Summary1 session · 1 premium request · Last updated 1 minute ago

Thanks for asking me to work on this. I will get started on it and keep this PR's description up to date as I form a plan and make progress.

Original prompt

You can make Copilot smarter by setting up custom instructions, customizing its development environment and configuring Model Context Protocol (MCP) servers. Learn more [Copilot coding agent tips](#) in the docs.

Link

Extend using MCP





Coding agent features demo part one

Best practices

CODING AGENT

Well-scoped issues

Making sure your issues are well-scoped

Choosing tasks

Choosing the right type of tasks to give to Copilot

Use comments

Using comments to iterate on a pull request

Custom instructions

Adding custom instructions to your repository

Extend capabilities

Using the Model Context Protocol (MCP)

Pre-install dependencies

Pre-installing dependencies in GitHub Copilot's environment





Issues are well-scoped

- A clear description of the problem to be solved or the work required.
- Complete acceptance criteria on what a good solution looks like (for example, should there be unit tests?).
- Directions about which files need to be changed.



Choosing the right type of tasks



**Complex and broadly
scoped tasks**



**Sensitive and critical
tasks**



Ambiguous tasks



Learning tasks



Comments on a pull request

components/ui/FlipNumber.tsx Outdated

9 9 `onPress?: () => void`

10 + `testID?: string`

 **timrogers** 8 hours ago

Let's make the `testID` mandatory so it's easier to write automated tests



 **Copilot** AI 8 hours ago

I've made the `testID` prop mandatory as requested. This will ensure all instances of the `FlipNum` component have a `testID`, making it easier to write automated tests. The existing usage already inc so there were no breaking changes. Commit: [b388336](#)

Help improve Copilot by leaving feedback using the 👍 or 👎 buttons

 Reply...

Resolve conversation



Custom instructions

is a Go based repository with a Ruby client for certain API endpoints. It is primarily responsible for ingesting metered usage for GitHub and recording that usage. Please follow these guidelines when contributing:

Code Standards

Required Before Each Commit

- Run 'make fmt' before committing any changes to ensure proper code formatting
- This will run gofmt on all Go files to maintain consistent style

Development Flow

- Build: 'make build'
- Test: 'make test'
- Full CI check: 'make ci' (includes build, fmt, lint, test)

Repository Structure

- 'cmd/': Main service entry points and executables
- 'internal/': Logic related to interactions with other GitHub services
- 'lib/': Core Go packages for billing logic
- 'admin/': Admin interface components
- 'config/': Configuration files and templates
- 'docs/': Documentation
- 'proto/': Protocol buffer definitions. Run 'make proto' after making updates here.
- 'ruby/': Ruby implementation components. Updates to this folder should include incrementing this version file using semantic versioning: 'ruby/lib/billing-platform/version.rb'
- 'testing/': Test helpers and fixtures

Key Guidelines

1. Follow Go best practices and idiomatic patterns
 2. Maintain existing code structure and organization
 3. Use dependency injection patterns where appropriate
 4. Write unit tests for new functionality. Use table-driven unit tests when possible.
- Document public APIs and complex logic. Suggest changes to the 'docs/' folder appropriate



Extend coding agent with MCP

MCP provides a standardized way to connect AI models to different data sources and tools, enabling them to work together more effectively. Below are some best practices

Using MCP

Enabling third-party

MCP servers for use may impact the performance of the agent and the quality of the outputs.

Check write access

Update the tools field in the MCP configuration with only the necessary tooling.

Carefully review

Carefully review the configured MCP servers prior to saving the configuration to ensure the correct servers are configured for use.



Pre-install dependencies in GitHub Copilot environment

- Powered by GitHub Actions, Copilot has access to its own ephemeral development environment
- Configure a *copilot-setup-steps.yml* file to pre-install these dependencies



Coding agent best practices demo part two



Questions?



Thank you